

CALCULUS I, QUIZ-2, 24/02/26 Name: TEJAS SAM

- (4 points) Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be a function such that $|f(x)| \leq |x|^2$ for all $x \in \mathbb{R}^n$. Show that f is differentiable at $(0, \dots, 0)$.
- (3 points) Identify the set M_2 of real 2×2 matrices with $\mathbb{R}^4$ using the identification:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \mapsto (a, b, c, d).$$

Consider the map $f: M_2 \rightarrow \mathbb{R}$ given by $f(A) = \text{trace of } A^2$. Find the derivative of f at the identity matrix.

- (3 points) Prove that there is no differentiable function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ such that $D_u(f)(a) > 0$ for a fixed $a \in \mathbb{R}^n$ and every nonzero vector $u \in \mathbb{R}^n$.

Q1: $|f(x)| \leq |x|^2, \forall x \in \mathbb{R}^n$, to show f is differentiable at $(0, \dots, 0)$

$$\hookrightarrow |f(0, \dots, 0)| \leq |0, \dots, 0|^2 = 0 \Rightarrow f(0, \dots, 0) = 0$$

$\rightarrow f$ is differentiable at $(0, \dots, 0)$ if $\exists A$, s.t. such that

$$\lim_{h \rightarrow 0} \frac{|f(0, \dots, 0+h) - f(0, \dots, 0) - A h|}{|h|} = 0$$

$\rightarrow$ Guess $A = 0$ mat

$$\text{then, } \lim_{h \rightarrow 0} \frac{|f(0, \dots, 0+h) - f(0, \dots, 0) - A h|}{|h|} = \lim_{h \rightarrow 0} \frac{|f(h) - 0 - 0|}{|h|} \leq \lim_{h \rightarrow 0} \frac{|h|^2}{|h|}$$

$$\therefore \lim_{h \rightarrow 0} \frac{|f(h) - f(0) - 0 \cdot h|}{|h|} \leq \lim_{h \rightarrow 0} \frac{|h|^2}{|h|} = \lim_{h \rightarrow 0} |h| = 0$$

$\therefore f$ is differentiable at $(0, 0, \dots, 0)$, and $f'(0, \dots, 0) = 0$ mat.